

SFMA TREATMENT GUIDELINES



There are three concepts that are important to discuss before going through all the treatment options available to the SFMA clinician. These concepts are critical in understanding how clients progress through each phase of treatment and how interventions are selected.

CONCEPT 1: FUNCTIONAL CONTINUUM

Most patients will enter your clinic with an obvious source of pain or dysfunction - be it a painful lower back, swollen knee or a restricted shoulder. Their true cause of pain or dysfunction is less obvious. That is the magic of the SFMA. It can clear away all the compensations and conscious sources of pain or dysfunction and highlight the true cause of the problem. Because of this, patients fall into one of four phases on a functional continuum:

- 1. Subconscious Dysfunction** - this is the initial phase for most patients when first seen by the clinician. This is where the patient is totally unaware of their true dysfunction (it is in their subconscious) or they are convinced their problem is elsewhere.
- 2. Conscious Dysfunction** - this is what happens after you perform an SFMA. The patient is now aware of their true dysfunction (it is in their consciousness) and they can start to address the real cause.
- 3. Conscious Function** - this phase is entered into once the patient can perform the correct functional pattern, but the pattern is not automatic (it is only functional with conscious control). They still need conscious effort to perform a good pattern of movement.
- 4. Subconscious Function** - the final stage is where the patient can now perform a functional pattern automatically (it is in their subconscious control) without having to think about the correction.

CONCEPT 2: THE THREE "R's"

There are different phases of treatment a patient can fall into depending on their cumulative SFMA findings. There are three treatment phases:

- 1. Reset** - This is defined by any manual intervention that helps the patient improve mobility or motor control with little or no active participation. This means a reset should not be a guided exercise. It should be a manual procedure demonstrated by competent clinical maneuvers. These can include mobilization, manipulation, chiropractic adjustment, myofascial work, soft tissue work, trigger point therapies, dry needling, acupuncture, muscle energy techniques, craniosacral techniques, A.R.T. (Active Release Technique), instrument assisted techniques, augmented therapy and any other techniques that involve kinesthetic input and manipulation or interaction with tissues.

It is easy to see how a reset can be done for the purposes of increasing mobility. However, many techniques also offer resetting in the chance that the patient does not have mobility problems but has motor control problems. The two simplest techniques to increase sensory input and manually heighten sensory awareness in situations of poor motor control, poor balance and poor stability are seen in techniques that involve both distraction of joints and compression of joints. Also, soft tissue compression, tactile cues, elongation of tissues and fascial interaction can actually improve

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sensory awareness and contribute to heightened motor control. These techniques are specific methodologies and require competency and training. Most clinicians are not expertly trained in all of these techniques but at least possess a few of these techniques that could form the criteria for reset.

2. Reinforce - These are techniques that are used to either maintain the reset that was just performed by the practitioner or help continue the reset while the patient is at home or in-between appointments. Reinforcement examples include stretching, postural aids, arch supports, taping, etc. This also includes how a clinician discusses with a patient the contraindications that have become necessary to enhance their rehabilitation. Simple discussions about hydration, sleeping position, ergonomic position, driving position, workout modification (so as not to exacerbate symptoms), footwear, orthotics, bracing, splinting and weight-bearing status as well as the use of forms of biofeedback, leukotape and kinesio tape create a proper reinforcement and behavior modification that facilitates the original manual treatment. This is done in an attempt to not have the patient default back to a poor or old movement pattern following a successful treatment simply by perpetuating poor habits and activities that could exacerbate pain and dysfunctional movement behavior.

3. Retrain - Lastly is reload. Retraining is the application of a motor learning / motor control exercise that is deduced from the SFMA particularly to improve a SMCD. It would be inappropriate to attack a true mobility dysfunction with a motor control exercises technique. It would be much better to try a manual mobility or stability technique first and demonstrate some degree of improvement through proper resetting of the sensory information followed by adequate reinforcement that together demonstrate improved movement behavior even if it only seems to be temporary. This is the window of opportunity where exercise finds its most fertile soil.

A Deeper Discussion of the 3 R's

As we review cases where the SFMA has been applied, we find the greatest success rate is from clinicians that intuitively and automatically create opportunities for improved sensory input in each of the three criteria previously stated. If, however, a clinician attempts to “skip steps” and move straight into reinforcement or retraining in the presence of a mobility dysfunction, less than optimal results are seen. Likewise, if a clinician continues to only provide manual resets for stability/motor control dysfunctions, versus moving through the 3 R's progression, they will also see less than optimal results.

Consider the following - offering a foot orthotic to an arch that has poor sensory stabilization and behavior without first creating more range of motion in ankle dorsiflexion will often perpetuate an uncomfortable situation. Remember, compensation usually causes poor motor control and poor motor behavior. It is human nature to give up stability at one segment when mobility at another segment is deficient. When dorsiflexion is limited at the ankle, the foot is obliged and obligated to pronate. The knee is obliged and obligated to have some degree of valgus collapse. The hips are obliged to perform differently and the foot intrinsics become less engaged all in an attempt to create freedom of movement as the body passes over the foot. If, however, successful manual work creates improved ankle mobility, then an opportunity for better stabilization elsewhere is much better received.

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A more appropriate course of action would be to complete a local examination of the ankle and foot to better decide if joint mobilization or soft tissue work, or both are needed. Once the proper joint mobilizations or soft tissue techniques are applied (reset), we must retest to verify a change in range of motion did actually occur. If there is an increase in ankle dorsiflexion range of motion it is appropriate to utilize some degree of taping, and/or stretching, to reinforce motion the body is adjusting to. When adequate range is restored, proper stability/motor control retraining is necessary - the body needs to learn how to properly use the motion. This step is vital in changing movement patterns - we need to reestablish proper motor control of a motion that has not been used.

Likewise, cervical range of motion may be temporarily enhanced with the introduction of tape across certain muscles of the neck and shoulder, but if greater range of motion can first be established through soft tissue or manual therapy techniques, then the tape need not be the therapy. It can simply reinforce the increased proper motion over the next few days to provide subtle sensory feedback through its orientation and contact to improve awareness. The SFMA guides us to the proper start point for our interventions.

In a situation where the patient has demonstrated full mobility, it is not necessary to start with the reset step. This patient would benefit from directed motor control retraining to improve their movement patterns. It is important to note, if reloading is used on an area that does not have adequate range of motion, you will be simply creating a new bad motor pattern, a new compensation.

This is also a good time to reiterate how the SFMA model for corrective exercise is more efficient and more aggressive than sending patients home with a few corrective exercises. Remember, the SFMA model attacks the dysfunctional non-painful pattern and is far less likely to have the home exercise program exacerbate symptoms because we are not working or expecting improvement on a painful pattern. We realize that pain compromises motor control when any attempt for self-administered home exercise would probably create a problem when expecting appropriate motor control to resume.

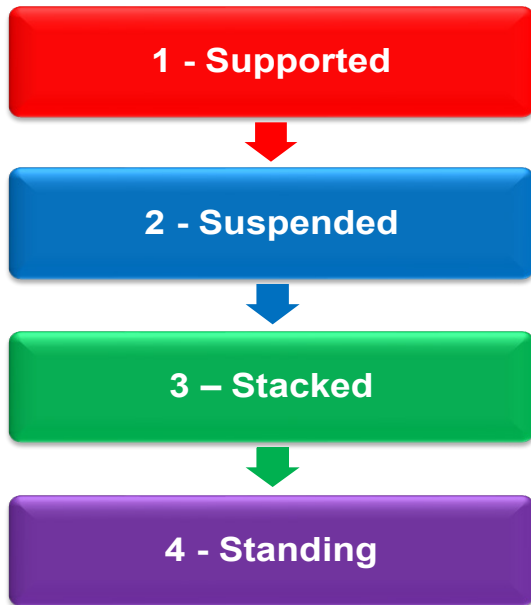
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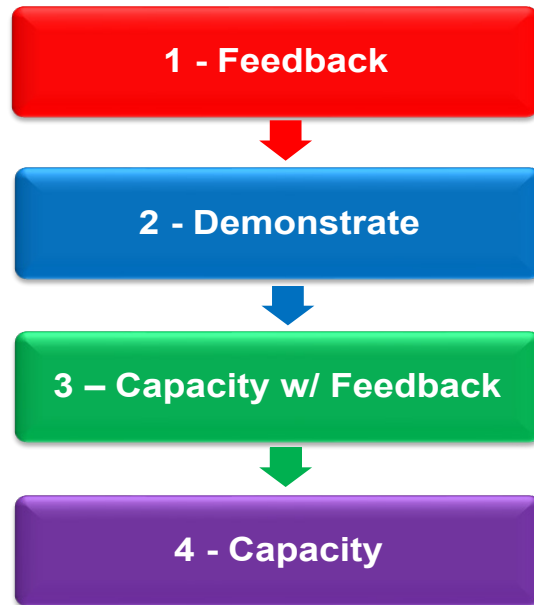
CONCEPT 3: 4 x 4 Matrix

This last concept describes the matrix to follow when naming and building the SFMA exercise progressions. This is how we systematically retrain new software into a dysfunctional system. “4 x 4” refers to the four possible exercise positions, combined with the four phases of challenges used.

The Four Postions:



The Four Phases of Challenges:



We can increase the difficulty of any exercise by either:

- changing the position - supported being the easiest and standing being the toughest
- changing the resistance - feedback being the easiest and capacity being the hardest

All exercises can be classified into one of these combinations. For example, a Half-Kneeling Chop is a (3-4) because it demonstrates Capacity in a Stacked Spine and Hip position. The easiest exercise would be a (1-1) Supported Posture with Feedback. An example would be Assisted Supine to Prone Rolling. The most difficult exercise would be a (4-4) Standing while Demonstrating Capacity exercise.

What is Feedback?

Feedback is defined as any technique to inhibit or facilitate tone, assist a movement or alter a motor control firing sequence during or before a task that is required. Or in another word, feedback is anything which helps the patient complete the given task correctly. A patient should demonstrate appropriate control before progressing to activities without feedback or before attempting to add capacity. Taking these steps to help develop subconscious function of motor control can be the difference between success and failure.

The most common progression to follow for the 4X4 matrix is the following:

1-1; 1-2; 2-1; 2-2; 3-1; 3-2; 4-1; 4-2; 1-3; 1-4; 2-3; 2-4; 3-3; 3-4; 4-3; 4-4.

MOBILITY GUIDELINES



All Mobility Treatment Boxes

As with all treatment boxes, before initiating treatment, perform any pertinent special ortho-neurological testing for the neuro-musculoskeletal system. Also, consider any other related differential diagnoses such as diabetes, vascular disorders, etc.

All mobility dysfunction boxes have the following recommended corrective strategies:

1. Perform local biomechanical and tissue testing to identify appropriate treatment protocol.
2. Start with appropriate mobility reset techniques.
3. Provide re-enforcement protocols whenever possible.
4. Once mobility is established, treat as if it were a SMCD.

BASIC INTERVENTION EXAMPLE - SPINE FLEXION



1 X 1 Exercises:

Supine Curl Up with Pattern Assistance:



Supine Oblique Curl Up with Pattern Assistance:



Assisted Supine to Prone Upper Body Rolling:



1 X 2 Exercises:

Supine Curl Up:



Supine Oblique Curl Up:



BASIC INTERVENTION EXAMPLE - SPINE FLEXION



2 X 1 Exercises:

Quadruped Reachunders (ER) with Feedback:



Targeted Cats and Dogs with Feedback:



2 X 2 Exercises:

Quadruped Reachunders (ER):



Targeted Cats and Dogs:



BASIC INTERVENTION EXAMPLE - SPINE FLEXION



3 X 1 Exercises:

Half-Kneeling Trunk Curls with feedback:



Half-Kneeling Oblique Trunk Curls with feedback:



3 X 2 Exercises:

Half-Kneeling Trunk Curls:



Half-Kneeling Oblique Trunk Curls:



BASIC INTERVENTION EXAMPLE - SPINE FLEXION



4 X 1 Exercises:

Toe Touch Progression -Heels Up and Down with Pattern Assistance



4 X 2 Exercises:

Reverse Toe Touch:

